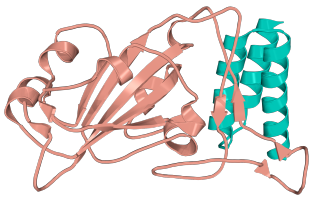
Biodefence by Design

Computational Biology for the Next Pandemic

BIO × AI Hack 2026 | AminoAnalytica Track

Dates:	1 – 8 March 2026
Venue:	Imperial College London, South Kensington
Format:	Teams of up to 5
Prizes:	£250 cash + Amina platform credits
Workshop:	Thursday 5th March, 6:00 PM



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The Challenge

The world got lucky with COVID-19. The pathogen that shut down civilisation had a case fatality rate below 2%. Next time, we might not be so fortunate.

Right now, H5N1 avian influenza is circulating across more species and more continents than at any point in history. It has jumped from birds to cattle, from cattle to humans, and is acquiring mutations that increase its affinity for human receptors. Scientists at the University of Glasgow have described the situation as "completely out of control." Meanwhile, the computational tools that could help us respond have never been more powerful.

Your challenge: leverage the tools of modern computational biology to build a solution that strengthens our ability to prepare for, detect, or respond to biological threats, whether natural or engineered.

The problem space is deliberately broad — we want you to find the question that excites you. But the deliverable must be specific. We are not looking for general-purpose pipelines or theoretical frameworks. We want you to pick a concrete biological target or threat, apply your approach to it, and show us the results. A binder designed against a named pathogen with predicted affinities is worth more to us than an elegant pipeline with no worked example. Choose a target, do the work, and prove it works!

Why This Matters Now

The convergence of AI and biology is one of the most consequential developments of our time. Tools like AlphaFold 3, RFdiffusion3, Boltz-2, and ProteinMPNN can now design functional proteins from scratch (antibodies, diagnostics, therapeutics) in hours rather than years. The first computationally designed protein medicine, SKYCovione (a COVID-19 vaccine developed at the UW Institute for Protein Design), has been approved in South Korea, the UK, and by the WHO. De novo antibody design was demonstrated with atomic-level accuracy for the first time in 2025.

But every tool that can design a therapeutic can, in principle, design a threat. In October 2025, a paper in *Science* demonstrated that open-source protein design models could redesign known toxin sequences such that they retained biological function while evading the biosecurity screening software used by every major DNA synthesis provider. The redesigned sequences folded into functionally identical structures, yet the screening tools simply missed them. RFdiffusion3 generates proteins that satisfy the constraints you give it, regardless of whether the target is therapeutic or dangerous. As these tools become more accessible, the gap between what a well-resourced lab can do and what a single individual with a laptop can do continues to narrow.

The problem extends beyond protein design. Large language models can now plan multi-step experimental workflows and interface directly with cloud laboratories via API. A recent preprint from NIST and Microsoft found that while current AI protein design tools cannot yet reliably produce functional harmful agents, each generation of models closes the gap.

This creates an urgent need for people who understand the technology deeply enough to build countermeasures at the same pace the capabilities advance: better structural screening algorithms, anomaly detection systems, governance frameworks, and forensic attribution tools. These are engineering problems, and they need engineers. The hackathon projects that tackle these challenges head-on are exactly the kind of work the field needs most.

AminoAnalytica works at this intersection. We partner with ARIA (the UK's Advanced Research and Invention Agency) on biosecurity research, and our AI-designed therapeutics have been clinically validated in partnership with biotech companies. Real compounds, treating real

patients. We built this track because we believe the next generation of biosecurity researchers will come from communities exactly like this one.

Suggested Directions

You are free to pursue any project at the intersection of computational biology and biosecurity. To get you started, here are some directions we think are especially compelling. If you have trained your own models or developed novel approaches as part of your research, you are welcome to bring those alongside the provided tools.

Important: in every case, we expect you to select a specific biological target and demonstrate your approach against it with concrete computational results. Do not submit a general-purpose tool without a worked application. The target you choose and the depth of your analysis on that target is what will set your submission apart.

Binder Design: Diagnostics & Therapeutics

Easy

At its core, both rapid diagnostics and therapeutic countermeasures start from the same problem: designing a protein that binds a pathogen target with high affinity and specificity. The difference lies in what happens after binding.

For diagnostics, you need a binder that recognises any accessible region of a pathogen surface protein, something that can be incorporated into a lateral flow test or biosensor. Current lateral flow tests rely on antibodies that take 6–12 months to develop and validate, with only ∼6% of candidates passing rigorous screening. Miniproteins and computationally designed binders offer a radically faster alternative.

For therapeutics, the binder needs to do more. It must target a functionally important region and disrupt the pathogen's mechanism of action, for example blocking receptor binding, preventing membrane fusion, or neutralising a toxin. The ability to computationally design these kinds of protein-based therapeutics (neutralising binders, engineered antibodies, antiviral peptides) has been demonstrated repeatedly in the literature. The challenge is speed: can we go from target identification to a panel of validated candidates fast enough to matter during an outbreak?

Pick a specific pathogen target — H5N1 haemagglutinin, Nipah virus G protein, Ebola GP, SARS-CoV-2 spike, or another target you can justify — and design a panel of candidate binders against it. Use tools like RFdiffusion and ProteinMPNN to generate designs, validate them computationally with Boltz-2, and score predicted complexes using IPSAE (average ipSAE across the interface) to quantify binding confidence. Classify candidates by application (diagnostic vs. therapeutic) based on their binding site and predicted mechanism. We want to see actual designed sequences with predicted binding affinities for your chosen target.

Broad-Spectrum Countermeasure Design

Medium

A binder designed against one strain of influenza may be useless against the next. The most valuable countermeasures are the ones that work across variants, or even across related pathogen families, by targeting conserved structural features that the pathogen cannot easily mutate away from.

This is a protein engineering challenge at its core. Using structural alignment and conservation analysis, you can identify epitopes that are shared across multiple strains or species. You can then use RFdiffusion and ProteinMPNN to design binders that target these conserved regions, and validate cross-reactivity by predicting binding affinity against multiple variant structures with Boltz-2 or molecular docking. The goal is a single designed protein (or a small panel)

that retains activity across a broad threat landscape, rather than being locked to one specific sequence.

Pick a specific rare or emerging disease — Nipah virus, Marburg, Lassa fever, a novel influenza subtype, or another pathogen family of your choice — and design a binder or therapeutic candidate that targets structurally conserved epitopes. Then validate it: run your designs against multiple named variant structures and present predicted binding affinities for each.

Biosecurity Screening & Safeguards

Hard

Current DNA synthesis screening operates at the sequence level, checking orders against databases of known threat agents. As outlined above, AI-redesigned sequences can retain dangerous function while evading these checks entirely. The field needs computational filters that catch dangerous designs before they reach a laboratory.

Several approaches are worth exploring. Structural homology screening would predict the fold of a designed protein and compare it against databases of known toxins, virulence factors, and immune evasion proteins, catching threats that sequence-level tools miss. Motif-level analysis would flag conserved dangerous features: receptor-binding domains, membrane-disrupting regions, protease cleavage sites, or aggregation-prone sequences associated with misfolded proteins like prions, particularly when designs show high structural homology to disease-causing pathogens despite low sequence identity. Behavioural anomaly detection would classify usage patterns themselves, distinguishing a researcher designing diagnostic binders from someone systematically optimising for immune evasion.

Pick a specific class of threat — redesigned toxins, immune evasion proteins, prion-like sequences, or another category you can define clearly — and demonstrate your screening approach against it. Assemble a test set of known dangerous sequences alongside benign controls, run them through your tool, and present the results: detection rates, false positive rates, and specific examples of what your approach catches that sequence-level screening would miss. We want evidence that your method works on real cases.

Novel Approaches

We explicitly encourage projects that do not fit neatly into the categories above. If you have an idea that uses computational biology to make the world more resilient to biological threats, we want to see it. Enzyme engineering for decontamination, phage-based countermeasures, population-level modelling, synthetic biology circuit design for biosensors: surprise us.

Tools & Resources

The Amina Platform

All teams on this track will receive access to **Amina**, AminoAnalytica's platform for protein design and analysis. Amina lets you run 50+ computational biology tools with remote GPU compute, without the hassle of setting up your own infrastructure. The toolkit spans structure prediction (Boltz-2, ESMFold, OpenFold3, Chai-1), protein design (RFdiffusion, ProteinMPNN, LigandMPNN), molecular docking (DiffDock, AutoDock Vina), molecular dynamics (OpenMM), property prediction (solubility, toxicity, aggregation, stability), and more.

There are three ways to use the platform: the CLI, the browser-based Toolkit, and the Chat interface. For this hackathon, we strongly recommend the CLI. It is the most powerful way to interact with Amina and the one that unlocks the most interesting workflows.

Getting Set Up

You will need Python 3.11+ installed. Setup takes about two minutes.

Amina CLI Setup

1. **Install the CLI:**
   ```
   pip install amina-cli
   ```
2. **Authenticate** (get your API key at https://app.aminoanalytica.com/settings/api):
   ```
   amina auth set-key "ami_your_api_key"
   ```
3. **Explore the toolkit:**
   ```
   amina tools
   ```
   ```
   amina tools -category design
   ```
   ```
   amina tools -search dock
   ```
4. **Run your first tool:**
   ```
   amina run esmfold -sequence "MKFLIL..." -o ./results/
   ```
5. **View results** in your local output folder or on the Jobs Dashboard.

Every tool has built-in documentation. Run `amina run <tool> -help` to see parameters, defaults, and usage tips. For long-running jobs, add the `-background` flag to free up your terminal, then check status with `amina jobs status <job_id>` and download results with `amina jobs download <job_id> -o ./results/`.

API keys and compute credits will be distributed at the opening ceremony and at the Thursday workshop. Full documentation is available at https://aminoanalytica.com.

Agentic Workflows: Claude Code, OpenClaw, and Amina

This is where things get interesting. The Amina CLI is designed to be driven by AI agents, and two of this event's sponsors provide exactly that capability.

Claude Code is Anthropic's agentic coding tool that runs directly in your terminal alongside the Amina CLI. When you combine the two, you get something close to an autonomous scientist: an AI agent that can plan multi-step protein engineering workflows, call Amina tools, interpret results, iterate on designs, and produce a final report, all from a single natural language instruction. This is our recommended setup for this hackathon.

To set this up, install the Amina Skills package in your project directory:

Agent Integration Setup (Claude Code)

```
npx skills add AminoAnalytica/amina-skills -all
```

Then run the `/amina-init` command inside Claude Code (or Cursor, Windsurf, or any compatible agent) to initialise. The skills teach the agent how to discover tools, format commands, chain outputs, and handle errors.

OpenClaw is an open-source AI agent framework that is also sponsoring this event. OpenClaw is model-agnostic (it works with Claude, GPT, Gemini, and others) and extensible through a

modular skills system. Unlike Claude Code, which runs as a CLI session, OpenClaw runs as a persistent background agent that can be reached through messaging platforms like WhatsApp, Telegram, and Slack. It can execute shell commands, manage files, and chain autonomous actions, which means it can orchestrate Amina CLI workflows in the same way Claude Code can. If you want to build a protein engineering agent that runs continuously, monitors results, and messages you when a design pipeline completes, OpenClaw is worth exploring.

You could also combine the two: use Claude Code for interactive design sessions where you are actively steering the workflow, and OpenClaw for longer-running autonomous tasks that you want to fire and forget.

Z.ai will additionally be providing LLM credits to all participants, so you will have access to frontier model compute for powering these agentic workflows.

Once configured, you can work entirely in natural language. The agent will translate your objectives into CLI calls, manage file I/O, and adapt when things do not go as expected.

Example: Agent-Driven Binder Design

```
> "Use the Amina CLI to run RFdiffusion against PDB 7TYK chain A, generate
50 backbones, pipe them through ProteinMPNN for sequence design, then
validate the top 10 with Boltz-2.  Score by predicted binding affinity
and present results."
```

The agent will break this into discrete CLI calls, execute them, parse the outputs, and reason about which designs to advance, all autonomously.

This is the frontier of autonomous scientific research. Real research groups are beginning to deploy exactly this kind of agent-driven workflow. Building one during a hackathon, and applying it to a genuine biosecurity challenge, is a serious achievement. We will be paying close attention to teams that integrate agentic approaches into their submissions.

Other Resources

You are welcome (and encouraged) to use any additional tools, datasets, or computational resources you have access to. This includes but is not limited to:

- Public databases: PDB, UniProt, GISAID, NCBI GenBank
- Open-source tools: AlphaFold, ESM-2, ColabFold, PyMOL
- Imperial HPC resources (if you have access)
- Any programming language, framework, or library

Workshop: Thursday 5th March, 6:00 PM

We will run a hands-on workshop introducing the Amina platform and the key computational biology tools relevant to this track. The session will cover:

- Getting started with Amina: accounts, credits, and basic workflows
- Protein structure prediction and interpretation
- De novo binder design: from target to candidate
- Tips for building a strong hackathon submission

Whether you are a seasoned computational biologist or an AI/CS student who has never touched a protein, this workshop will get you up and running. No prior experience required.

Judging Criteria

Submissions will be evaluated by the AminoAnalytica team on the following criteria:

Criterion	Description	Weight
Impact	Does this address a real biosecurity need? How significant is the problem, and how meaningful is the proposed solution? Projects that demonstrate results against a specific, named biological target will be scored higher than general-purpose tools without a worked example.	30%
Scientific Rigour	Is the computational approach sound? Are results validated appropriately? Are limitations acknowledged?	25%
Technical Execution	How well-built is the solution? Does it work? Is the code or pipeline reproducible?	20%
Creativity	Is the approach novel or surprising? Does it combine tools or ideas in interesting ways?	15%
Presentation	Is the submission clearly communicated? Could someone outside your team understand and build on your work?	10%

Submission Requirements

Teams should submit the following via the DoraHacks platform:

- A **project write-up** (max 4 pages) describing your problem, approach, results, and next steps
- **Code, pipelines, or notebooks** used in your project (GitHub repo or equivalent)
- A **short demo or presentation** (3–5 minutes) for judging day
- Any **designed sequences, structures, or datasets** your project produced

A note on what we are looking for: every submission must include a specific biological target or threat, the rationale for choosing it, and concrete computational results demonstrating that your approach works on that target. A well-executed project against a single pathogen will always be valued more highly than an ambitious framework with no worked application.

Prizes

Place	Prize
1st Place	£125 + Amina platform credits
2nd Place	£75 + Amina platform credits
3rd Place	£50 + Amina platform credits

Outstanding projects may also be invited to continue their work with AminoAnalytica, including the possibility of experimental validation of computationally designed candidates.

Get Started

1. **Form a team** (1–5 people, any background welcome)
2. **Pick a specific target** — a pathogen, a threat class, a concrete biosecurity problem
3. **Attend the workshop** on Thursday to get set up with Amina
4. **Build something real** — designs, screens, validated results against your target
5. **Submit** by Saturday 8th March

The next pandemic will not wait for us to be ready.

The question is whether we can build the tools, systems, and knowledge to respond before it arrives. That work starts here.

Questions? Reach out to founders@aminoanalytica.com

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https://aminoanalytica.com